

HOMEWORK ASSIGNMENT 10

CYCLE ANALYSIS

1. (10 points) Describe the difference between the Brayton cycle and the Rankine cycle, and again the difference between the Brayton cycle and the Otto cycle. Comment on both the working fluid and the structure of the systems that run the cycles.
2. (30 points) A vapor compression cycle uses R-134a as a refrigerant with a mass flow of 0.1 kg/s and runs between pressures of 0.16 MPa and 1.8 MPa. State 1 is a saturated vapor, and State 3 is a saturated liquid. The compressor has an isentropic efficiency of 90%. Use the NIST webbook to find the state points for R-134a.
 - a) What is the power required to run this cycle?
 - b) How much heat is extracted from the cold space?
 - c) What is the coefficient of performance of the cycle? How does this compare to the Carnot COP?
 - d) In an attempt to extract some work out of the cycle, you replace the throttle with a little turbine. Assuming the turbine is ideal and the pressure ratio across the turbine is the same as that of the throttle, how much work could you extract from the fluid? How does this change the heat extracted from the cold space assuming that State 1 is the same?
3. (30 points) The Taurus 60 gas turbine in the West Campus Steam Plant has a compressor pressure ratio of 12.2 and a mass flow rate of 21.5 kg/s. The combustor adds 18 MW of heat to the cycle, and the isentropic efficiency of the compressor and turbine are 0.85 and 0.9, respectively. Assuming an inlet condition of $T_1=278$ K and $P_1=95,900$ Pa to do an air-standard cycle analysis to calculate the following. Use $c_p=1001$ J/kg-K and $\gamma=1.4$.
 - a) Temperature and pressure at all four state points
 - b) Power produced by the engine
 - c) Engine thermal efficiency
4. (30 points) In this problem, you will compare the Otto and Diesel cycles for a few different conditions to help understand the differences between the two cycles and how they're typically used in piston engines.

Air standard, ideal Otto cycle:	Air standard, ideal Diesel cycle:
• Adiabatic, reversible compression • <i>Isochoric</i> heat addition • Adiabatic, reversible expansion • <i>Isochoric</i> heat rejection	• Adiabatic, reversible compression • <i>Isobaric</i> heat addition • Adiabatic, reversible expansion • <i>Isochoric</i> heat rejection

In all cases, you will assume the intake conditions (state 1) to be the following:

- $T_1 = 300 \text{ K}$
- $P_1 = 0.1 \text{ MPa}$
- All cases have a bottom dead-center volume (V_1) of 0.625 L.
- Assume air to have the following properties: $\gamma=1.4$, $c_v=714 \text{ J/kgK}$, $c_p=1000 \text{ J/kgK}$

Additionally, you will use a heat addition of 1500 J for all analyses.

- a. Analyze Otto and Diesel cycles for a compression ratio of 11 (high compression ratio gasoline engine, low compression ratio diesel). Fill out the following table for these two cycles:

Otto Cycle			
State	P [MPa]	T [K]	V [m ³]
1			
2			
3			
4			

Diesel Cycle			
State	P [MPa]	T [K]	V [m ³]
1			
2			
3			
4			

- b. Draw the P-v diagram for both cycles on the same plot (to scale as best you can). Show the net work on both and indicate which cycle has higher net work.
- c. What is the efficiency of each of these cycles?